

A CASE REPORT OF EXTENSIVE MULTI-SINUS OSTEOLYSIS CAUSED BY *ASPERGILLUS NIGER* FUNGAL BALL

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INTRODUCTION :

To highlight a rare presentation of non-invasive *Aspergillus niger* fungal ball causing extensive spontaneous sinonasal osteolysis and bilateral “auto-FESS” changes, mimicking invasive fungal rhinosinusitis in diabetic patient.

CASE PRESENTATION :

A 68-year-old diabetic woman who presented with chronic headache and foul-smelling nasal discharge for 6 months. Diagnostic evaluation included nasal endoscopy and contrast-enhanced computed tomography of the paranasal sinuses. She underwent bilateral functional endoscopic sinus surgery, followed by histopathological examination and fungal culture of intraoperative specimens.

DISCUSSION :

Radiological and intraoperative findings revealed extensive, smooth, well-remodelled bone resorption involving the medial maxillary walls, inferior ethmoid floors, and osteomeatal complexes, producing spontaneous bilateral antrostomies resembling previous functional endoscopic sinus surgery. The orbits were preserved. Complete spontaneous resorption of the right uncinate process and bulla ethmoidalis was noted, while a thin residual uncinate process remained on the left. Histopathology demonstrated strictly extramucosal septate fungal hyphae with calcium oxalate crystals, without mucosal, vascular, or osseous invasion, thereby excluding invasive fungal rhinosinusitis. Fungal culture yielded isolated *Aspergillus niger*.

This microbiological profile underpins a biochemically-mediated osteolytic process rather than mechanical or infectious destruction. *Aspergillus niger* secretes oxalic acid, which chelates local calcium to precipitate the observed calcium oxalate crystals. The resultant localized tissue acidosis drives direct chemical resorption of the surrounding osseous walls. This unique chemical pathway accounts for the extensive, smooth "auto-FESS" remodeling without invasion.

Complete mechanical clearance and high-volume saline irrigation resulted in full clinical resolution, with well- epithelialized cavities and no recurrence in 12 months follow-up.

CONCLUSION :

Non-invasive *Aspergillus niger* fungal ball can rarely cause extensive multi-sinus osteolysis, likely through localized chemical effects. Recognition of the smooth, well-remodelled “auto-FESS” pattern is essential to distinguish from the irregular, destructive pattern of invasive fungal rhinosinusitis. This distinction is particularly important in diabetic patients, as **accurate diagnosis may prevent unnecessary aggressive surgery and allow curative conservative endoscopic clearance.**

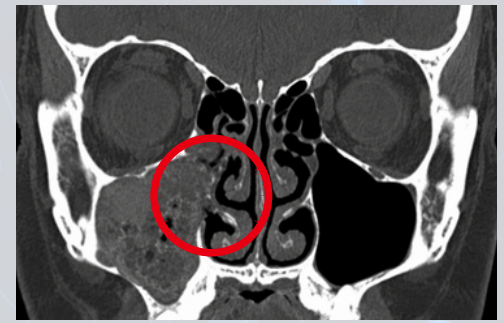


Figure 1 (axial view) & Figure 2 (coronal view): bone destruction and resorption of the medial wall of the both maxillary sinuses (red circle showed on right side), inferior wall of the both ethmoidal sinuses and osteomeatal complexes on CECT PNS.

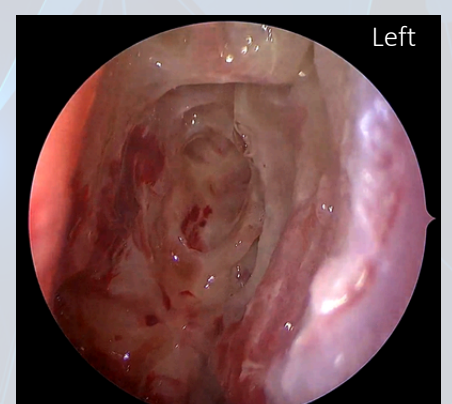
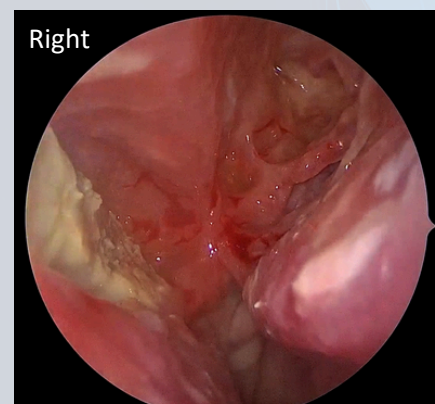


Figure 3 & 4 (intraoperative findings): Endoscopic appearance of both right and left sinonasal cavity demonstrated spontaneous enlargement of the maxillary sinus ostium with loss of normal osteomeatal complex architecture.

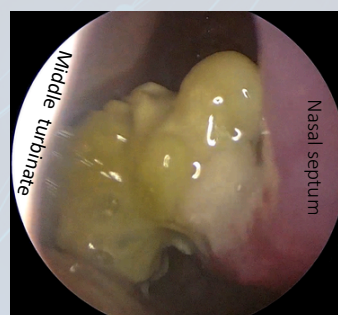


Figure 5: dense fungal material seen between nasal septum and medial turbinate (right side)

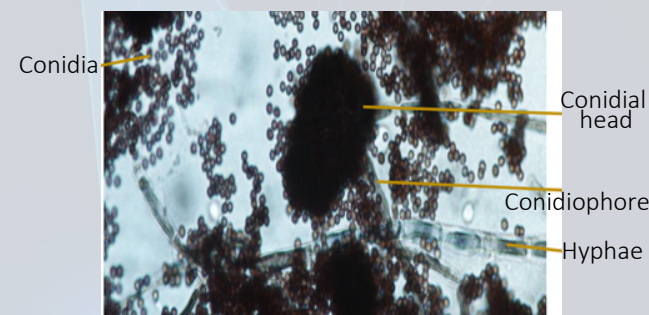


Figure 6: The colony exhibited a black surface with a peripheral white margin and cream-colored reverse. Microscopy showed hyaline septate hyphae, smooth-walled conidiophores, and dense globose conidial heads bearing rough-walled dark conidia, consistent with *Aspergillus niger*.

KEY MESSAGE :

- Rare presentation
- Smooth osteolysis may indicate non-invasive fungal ball.
- Recognize the “auto-FESS” sign
- Histopathology confirms the diagnosis

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DECLARATION OF CONFLICT OF INTEREST :

- All authors declare no conflicts of interest.
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